

Demonstration note — T1'→T4 analytic scheme (non-resolutive)

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Abstract

Caution. RH is open. This note demonstrates the *analytic scheme* T1'–T4: (T1') explicit formula (Guinand–Weil), (T2) spectral framing (Gram + Gershgorin/Schur), (T3) wave-packet localization, (T4) cross-positivity. We prove *the bounds* and *logic* of the scheme; we do not claim to resolve RH.

Preliminaries

Fourier: $\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-ix\xi} dx$, $\phi(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{\phi}(\xi) e^{ix\xi} d\xi$. Let $\mathcal{PW}$ be even, smooth, band-limited functions ($\text{supp } \widehat{\phi} \subset [-A, A]$).

T1' — Explicit formula (Guinand–Weil)

$$\sum_{\rho} \widehat{\phi}(\gamma_{\rho}) = \frac{1}{2\pi} \int K(t) \widehat{\phi}(t) dt - \sum_{n \geq 1} \frac{\Lambda(n)}{\sqrt{n}} (\phi(\log n) + \phi(-\log n)), \quad K(t) = 2 \log |t| + O(|t|^{-2}).$$

T2 — Spectral framing (Gram + Gershgorin/Schur)

Grid $\Xi_A = \{\xi_{p,k} = k \log p \leq A\}$, weights $w_{p,k} = \log p / p^{k/2}$, Gram $G_{\alpha\beta}^{(w)} = \sqrt{w_{\alpha}} \widehat{\phi}(\xi_{\alpha} - \xi_{\beta}) \sqrt{w_{\beta}}$, $R = G^{(w)} - I$. If $S := \sup_{\alpha} \sum_{\beta \neq \alpha} |R_{\alpha\beta}| < 1$ then $\lambda_{\min}(G^{(w)}) \geq 1 - S > 0$. Locality ($|\xi_{\alpha} - \xi_{\beta}| \leq A$) + p -adic decay $\Rightarrow S < 1$ for large A .

T3 — Wave-packet localization (contradiction in this scheme)

Assume an off-line zero $\rho_0 = \beta_0 + i\gamma_0$ ($\beta_0 \neq \frac{1}{2}$, $0 < \gamma_0 < A$). Build $\widehat{\phi}_T(\gamma) = \frac{1}{2}(b(\sqrt{T}(\gamma - \gamma_0)) + b(\sqrt{T}(\gamma + \gamma_0)))$, $\|\phi_T\|_2 = 1$. Then: Archimedean term $\rightarrow 0$ (DCT); prime-side $\rightarrow 0$ (IBP decay, finitely many $\xi_{p,k} \leq A$); zero-side $\rightarrow 1$ (frame capture). Plug into T1' $\Rightarrow$ contradiction (in this framing).

T4 — Cross-positivity (equality criterion)

$K_{\phi}(\rho, \rho') = \widehat{\phi}(\gamma - \gamma')$ is positive; so is the prime-side quadratic form. Cauchy–Schwarz yields a cross-inequality. Equality for *all* fine tapers forces critical symmetry.

We prove the *scheme* and its key bounds; we do not claim to settle RH.